Table 2. Comparison of temporal approaches in the fixed-parameter binary logit model of crash injury severity (serious injury/fatal vs. PDO/minor injury), Thailand, 2016–2023.

Temporal approach	Number of parameters	Variables with temporally varying parameters	Variables restricted to a proportional trend	LL(β)	AIC	BIC	LR test against the fully unconstrained model, χ^2 (df)	Confidence of rejection
S1: single-motorcycle crashes (two-year periods); N = 7,325; 40 candidate variables (including the constant)								
Fully unconstrained (separate-period logits)	160	all	–	–4,520.92	9,361.8	10,465.7	–	–
Partially constrained (LR tests, 90%)	73	26 of 40	–	–4,552.26	9,250.5	9,754.2	62.7 (87)	0.023
Spline indicator (90%)	78	40 of 40	12	–4,558.15	9,272.3	9,810.4	74.5 (82)	0.289
Adaptive fused (90% rule)	59	17 of 40	–	–4,579.68	9,277.4	**9,684.4**	117.5 (101)	0.875
S2: motorcycle and another vehicle type; N = 21,371; 46 candidate variables (including the constant)								
Fully unconstrained (separate-period logits)	368	all	–	–12,052.15	24,840.3	27,773.2	–	–
Partially constrained (LR tests, 90%)	109	38 of 46	–	–12,140.48	24,499.0	25,367.7	176.7 (259)	0.000
Spline indicator (90%)	143	46 of 46	4	–12,123.32	24,532.6	25,672.3	142.3 (225)	0.000
Adaptive fused (90% rule)	66	19 of 46	–	–12,218.17	24,568.3	**25,094.3**	332.0 (302)	0.887
S3: single vehicle, no motorcycle; N = 73,731; 51 candidate variables (including the constant)								
Fully unconstrained (separate-period logits)	408	all	–	–23,563.23	47,942.5	51,699.4	–	–
Partially constrained (LR tests, 90%)	103	32 of 51	–	–23,668.37	47,542.7	48,491.2	210.3 (305)	0.000
Spline indicator (90%)	135	51 of 51	8	–23,653.82	47,577.6	48,820.7	181.2 (273)	0.000
Adaptive fused (90% rule)	69	13 of 51	–	–23,748.79	47,635.6	**48,270.9**	371.1 (339)	0.889
S4: multi-vehicle, no motorcycle; N = 33,646; 45 candidate variables (including the constant)								
Fully unconstrained (separate-period logits)	360	all	–	–12,583.17	25,886.3	28,918.9	–	–
Partially constrained (LR tests, 90%)	99	31 of 45	–	–12,673.08	25,544.2	26,378.1	179.8 (261)	0.000
Spline indicator (90%)	137	45 of 45	5	–12,653.02	25,580.0	26,734.1	139.7 (223)	0.000
Adaptive fused (90% rule)	68	18 of 45	–	–12,743.54	25,623.1	**26,195.9**	320.7 (292)	0.881

Notes. Lowest BIC within each subset among models not rejected at the 90% level in bold. Fully unconstrained: a separate parameter for every variable in every period. Partially constrained: adjacent periods merged when a likelihood ratio test does not reject equal parameters at the 90% level. Spline indicator (Shabab et al., 2024): each variable enters through spline terms, the base term giving a parameter proportional to the period index; the other terms removed by backward elimination at the 90% level. Adaptive fused: sparsest penalty whose structure is not rejected at the 90% level against the fully unconstrained model (LR < $\chi^2_{0.90}$(df)). Variables with temporally varying parameters count variables (including the constant) whose parameter differs between at least two periods; for the spline every variable varies over time. LR test: $\chi^2 = -2[LL(\beta)_{model} - LL(\beta)_{fully\ unconstrained}]$, df = difference in the number of parameters; confidence of rejection below 0.90 means that the model is not rejected at the 90% level.